

# Zhang Yichen

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## EDUCATION

### Tsinghua University

Ph.D. Candidate in Chemical Engineering, GPA: 3.99/4.00, Ranking: 3/86 (Advisor: Assoc. Prof. Xuebing Zhao)

June 2023 - Present

### Xiamen University

B.Eng. in Chemical Engineering, GPA: 3.82/4.00, Ranking: 1/42

Sep 2019 - June 2023

## SCHOLARSHIPS & AWARDS

Friends of Tsinghua – Shanghai Huayi Scholarship, THU (Oct 2025)

Best Poster Award, 22<sup>nd</sup> National Electrochemical Congress (Oct 2024)

Teng-Teng Scholarship and Gao-Yingshi Scholarship, THU (Oct 2024)

Future Scholars Scholarship, THU (Sep 2023)

National Scholarship, XMU (Nov 2021, 2022)

Maotai Scholarship, XMU (Nov 2020)

## SKILLS AND INTERESTS

|                         |                                                                                                                                                                                                                                                                                |
|-------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Interests</b>        | Experimental and theoretical studies on non-precious-metal ORR catalysts; structure–activity relationships of atomically dispersed sites; machine-learning interatomic potentials (MLPs) and active-learning workflows for accelerating first-principles catalysis simulations |
| <b>DFT / MD</b>         | VASP, CP2K, Gaussian, Material Studio, LAMMPS, GROMACS, ASE, pymatgen, Multiwfn, vaspkit, VESTA                                                                                                                                                                                |
| <b>ML potentials</b>    | MACE (fine-tuning), DeePMD-kit / DPA4 (active learning), NEP / GPUMD                                                                                                                                                                                                           |
| <b>Programming</b>      | Python (NumPy, PyTorch, ASE, pymatgen), Bash / SLURM, Git, MATLAB, C                                                                                                                                                                                                           |
| <b>Characterization</b> | XAS analysis (Demeter, FDMNES, larch), XRD (Jade), XPS (Advantage), SEM / TEM; CAD, SolidWorks                                                                                                                                                                                 |

## RESEARCH EXPERIENCE

### DFT-guided design of atomically dispersed electrocatalysts

2023 – Present

Mechanistic DFT (VASP / CP2K) for single-atom and atomic-pair catalysts in ORR, CO<sub>2</sub>RR and propylene epoxidation, in close collaboration with experimental groups (Publications 1–5): reaction pathways, free-energy diagrams, electronic-structure descriptors, and comparison with in situ spectroscopy.

### Machine-learning potentials for accelerated catalysis simulation

2025 – Present

Built an active-learning geometry-optimization loop (MLP pre-relaxation → DFT → on-the-fly fine-tuning): on a 75-atom magnetic Fe–N–C ORR catalyst, fine-tuned MACE cut DFT ionic steps from 128 to 66–73, and revealed that DPA4-Air pre-relaxation quenches the Fe spin state. Developed dataset-audit and committee active-learning protocols for MACE / DPA4 fine-tuning; MLP-driven MD of Li-ion electrolytes and molten salts.

### Computational workflow and software projects

2025 – Present

Solid programming background. Developed an adsorbate-placement tool (particle-swarm search with rigid-ring kinematics) for generating DFT initial structures, an atomic-structure visualization tool, and an LLM-agent toolkit that automates VASP / CP2K / LAMMPS input generation, HPC submission and result verification.

## RESEARCH PUBLICATION

‡ = co-first authorship; \* = corresponding.

1. Zhao, W.‡; **Zhang, Y.‡**; Zhu, X.; Ran, L.; Tong, W.\*; Wang, M.; Zhou, H.; Chen, S.\*; Farràs, P.\*; Qiu, X.\* Motif Editing for Reaction Pathway Switching From Formate to CO in Single-Atom Indium Catalysts via Adjacent Carbon Site Manipulation. *Adv. Funct. Mater.*, **2026**, 36, e76951.

**Overview:** Incorporating O into the In–N<sub>4</sub> Lewis interface edits the motif to In–N<sub>2</sub>O<sub>2</sub>, switching the CO<sub>2</sub>RR product from formate to CO with >90% CO selectivity across –0.51 to –1.10 V vs. RHE. In situ spectroscopy and DFT reveal that In–N<sub>2</sub>O<sub>2</sub> enriches the electron density of adjacent carbon sites, facilitating \*COOH adsorption and CO formation.

2. **Zhang, Y.**<sup>‡</sup>; Gao, D.<sup>‡</sup>; Ouyang, D.; Yan, B.; Zhao, X.\* Efficient harvesting of electricity, aromatic aldehydes and H<sub>2</sub> from lignin over nanoflower-like cobalt-based bifunctional electrocatalysts. *Green Chem.*, **2025**, 27, 8517–8531.

**Overview:** A self-powered system using nanoflower-like Co bifunctional catalysts (high-current electrodeposition on Ni foam) that simultaneously harvests electricity (196.1 mW cm<sup>–2</sup> peak power), aromatic aldehydes (2.90–3.34% yield) and H<sub>2</sub> (12.7 mL cm<sup>–2</sup>) from lignin; the direct lignin fuel cell drives depolymerization and water splitting without external power.

3. Qiu, Y.<sup>‡</sup>; **Zhang, Y.**<sup>‡</sup>; Zhang, R.; Huang, M.; Tan, K. B.; Zhan, G.; Li, Q.; Huang, J.; Fu, G.\* Internally hollow Cu<sub>2</sub>O nanoframes with the abundance of {110} facets enhance direct propylene epoxidation. *Nat. Commun.*, **2025**, 16, 7802.

**Overview:** Engineered Cu<sub>2</sub>O nanoframes via ligand-protected facet-etching achieve 0.18 mmol g<sub>cat</sub><sup>–1</sup> h<sup>–1</sup> PO formation rate with 83.8% selectivity at 175°C. The catalyst maintains performance >300 min, with mechanistic studies (DFT/characterization) confirming Cu<sub>2</sub>O {110} facets promote active oxygen species generation and selective epoxidation.

4. Ran, L.<sup>‡</sup>; **Zhang, Y.**<sup>‡</sup>; Tong, W.; Chen, L.; Wang, M.; Zhou, H.; Farràs, P.\*; Chen, S.\*; Qiu, X.\* Circumventing Radical Generation on Fe–V Atomic Pair Catalyst for Robust Oxygen Reduction and Zinc–Air Batteries. *Angew. Chem. Int. Ed.*, **2025**, 64, e202514542.

**Overview:** Established an ORR pathway-switching strategy using DFT-guided Fe<sub>1</sub>V<sub>1</sub>–NC atomic pair catalysts to circumvent ROS generation. The N<sub>2</sub>–Fe–N<sub>2</sub>–V–N<sub>2</sub> configuration enables direct O=O bond cleavage (side-on adsorption), achieving record stability (16 mV decay @ 50k cycles) and activity (E<sub>1/2</sub> = 0.89 V). Demonstrated in zinc–air batteries with 400 h durability at 10 mA·cm<sup>–2</sup>.

5. Qiu, Y.; **Zhang, Y.**; Xiao, S.; Tian, J.; Zhang, R.; Sun, L.; Tan, K. B.; Duh, Y.-S.; Li, Q.; Huang, J.; Fu, G.\* Low-coordinate Cu<sub>1</sub>–N<sub>2</sub> sites in single-atom catalyst for highly efficient propylene epoxidation with molecular oxygen. *Nat. Catal.*, under revision.

**Overview:** Developed a novel Cu<sub>1</sub>–N<sub>2</sub> single-atom catalyst for direct propylene epoxidation, achieving record PO formation rates. Advanced characterization and DFT studies revealed exclusive Cu–N coordination enhances electrophilic oxygen species formation, enabling selective C=C bond activation.

6. Liu, X.; **Zhang, Y.**; Li, Y.; Yan, B.; Zhao, X.\* Facile preparation of 2D nanomaterials by chelating agent-assisted electrodeposition for efficient electrochemical oxidation of biomass-derived aldehydes coupled with CO<sub>2</sub> reduction. *Appl. Catal. B: Environ.*, **2025**, 368, 125146.

7. Li, Y.; Ouyang, D.; Liu, X.; **Zhang, Y.**; Niu, Z.; Zhu, J. Y.; Pan, X.; Zhao, X.\* Self-powered electrochemical synthesis of hydrogen peroxide from air and lignin. *Energy Environ. Sci.*, **2025**, 18, 3633–3646.

8. Dai, Z.; Liu, X.; Liu, N.; **Zhang, Y.**; Zhao, X.\* Upgrading biomass derived furan aldehydes by coupled electrochemical conversion over silver-based electrocatalysts. *Chem. Eng. J.*, **2024**, 488, 151001.

9. Xiao, S.; Wei, C.; Li, R.; Qiu, Y.; Tan, K. B.; **Zhang, Y.**; Huang, J.\*; Li, Q. Supported AgCuCl Nanoclusters as Bimetal Catalysts for Propylene Epoxidation with Molecular Oxygen. *ACS Appl. Nano Mater.*, **2023**, 6(11), 9687–9696.

## SUMMARY

- Proficient in VASP, CP2K, Gaussian, LAMMPS and machine-learning potentials (MACE, DPA4, GPUMD), including fine-tuning and active-learning workflows
- Experienced in both experiment (synthesis, electrochemistry, XAS / XRD / SEM / TEM) and theory; skilled in communicating with experimental collaborators
- Solid programming background: Python (NumPy, PyTorch, ASE, pymatgen), Bash / SLURM, MATLAB and C; builds automated simulation workflows